Comparing Models for Categorizing Apartments

Paulien Wisselink (2552970), Dave Ebbelaar (2540971), Jerney van Schagen (2550577), Pam Scheer (2540039), Quinten van Langen (2557739)

Vrije Universiteit Amsterdam, The Netherlands, Group 46
p.e.wisselink@student.vu.nl, d.ebbelaar@student.vu.nl, j.s.a.van.schagen@student.vu.nl, p.scheer@student.vu.nl, q.i.van.langen@student.vu.nl

Abstract. In this paper, five classification models are being analyzed. We used a pre-processed Kaggle competition dataset. RentHop, an American renting website, shows visitors an interest level per apartment. The task was to determine what interest level new apartments have by analyzing five classification models. The five models are: Random Forest, Linear Discriminant Analysis, Classification and Regression Trees, K-nearest Neighbors and Support Vector Machines. All the models were trained and tested using R-studio. We found that the Random Forest model is the most accurate model and has the smallest error rate. Kappa supported this claim.

1 Introduction

"Machine learning is the subfield of computer science that gives computers the ability to learn without being explicitly programmed "[1]. It originated from the 1930's from Ronald A. Fisher [2] where Fisher took 50 samples of three flowers of the same family and identified four features of those flower. He invented a linear model to categorize these flowers. After Fisher, many scientists followed and machine learning arose. The evolution of machine learning was a result of "fundamental advances in statistics, probability theory, and functional analysis as well as from the exponential increase in computing capability"[3]. As in the recent years, machine learning is used by countless companies. For example, machine learning is used by Facebook in the way that it recognizes certain faces in tagged photo's. Facebook's Machine Learning team even invented an algorithm to look for suicidal people and offer them help [4].

RentHop is an American website which offers multiple apartments in New York for rent. Instead of letting customers browse through all the available apartments endlessly, the website wants to rank the apartments by interest level from high to medium to low. Furthermore, creating this ranking can also help to detect fraud and understanding renters'needs and preferences. [5] However, given the many different features the apartments posses it is very complicated to find a suitable way of ranking them. For example, what feature weighs more than another or what feature can be prevailing? Therefore the main research question discussed in this paper will be:

Which of the five proposed models predicts the interest level (high/medium/low) of a particular apartment best?

2 Data inspection & Preparation

2.1 The dataset

The data available on the Kaggle website contained a test and training set, a sample of the images of the listings and all the images of the listings. However, the photos were not used for this project. The test and training dataset included the following features:

- number of bathrooms
- number of bedrooms
- building id
- the description of the apartments
- a display address
- the features of the apartment
- listing id
- the longitude and latitude of the apartment
- the manager id (the manager who sells the apartment)
- the price of the apartment in US dollar
- the street address
- the links to the photos of the apartment

The training set also includes an indication of the interest level: low, medium or high. Since the test set does not contain this interest level feature and is only used for the Kaggle competition, this set was not used for this project.

To correctly read the training dataset into Rstudio some adjustments were made. After importing the dataset was first converted to a Tibble data frame. Tibbles are a modern take on data frames. They keep the features that have stood the test of time, and drop the features that used to be convenient but are now frustrating (i.e. converting character vectors to factors). The Tibble format allows for a more flexible workflow. The features 'photos'and 'features of the apartment'both contain a list of items which are hard to analyse on its own. Therefore these features are converted to display the length of the list resulting in two new features:

- photo_count
- feature_count

Now to further clean up the dataset a fill function was used to add fill for listings lacking any features.

2.2 Feature extraction

From all the features of the listings, the 25 most frequent were extracted and added to the dataset as new features. This was done by first cleaning up the feature list by removing irrelevant frequent words and stopwords resulting in the following 25 new features: *balcony, cats, construction, deck, dining, dishwasher, dogs, doorman, elevator, fee, fitness, floors, hardwood, Internet, laundry, nofeat, outdoor, pool, pre, roof, speed, swimming, terrace, unit, war.* After this the original feature column was removed from the dataset.

2.3 Creating numerical values

To further clean up the dataset some features had to be adjusted. First the building and manager id and street and display address were converted to an integer value. Then the date from the 'created'feature was converted so it would display the date in three new features: months, days and hours. The original 'created'column was removed afterwards. The description feature was also adjusted to displays the length of the description in words. The dataset now only contains numerical values and is ready for analysis.

2.4 Statistical analysis

By using the summary and plot functions in Rstudio the data was analyzed to check the distributions. Overall the data was normally distributed with a few exceptions. Namely: price, photo_count and feature_count. In table 1 the distribution of those features is represented. All these features have big outliers.

Table 1: Distributions of 3 features

	Min	1st Qu.	Median	Mean	3rd Qu.	Max.
Price	43	2500	3150	3830	4100	4490000
Photo_count	0	4	5	5,607	7	68
Feature_count	0	2	5	5,428	8	39

To compensate for this a log transformation was used.

2.5 Log transformation

To do a logarithmic transformation, a calculation is made with the log of each value in the dataset and use those transformed values rather than the raw data. Log transforms tend to have a major effect on distribution shape, and in visualizations can bring extreme outliers closer to the rest of the data so graphs are not stretched out as much. As example two histograms, in the figures 1 and 2, show the difference of the price feature before and after the log transiformation.

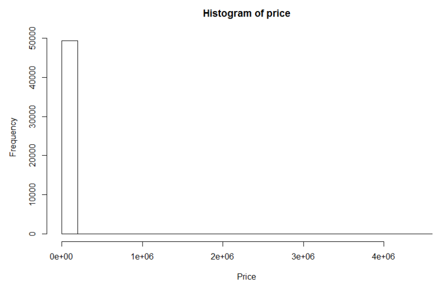

Fig. 1: Before log transformation

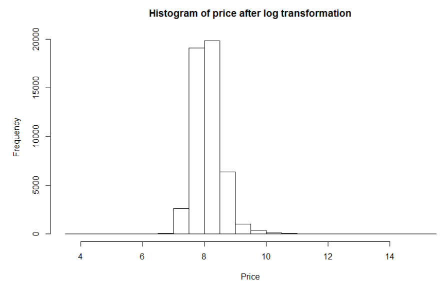

Fig. 2: After log transformation

2.6 Creating a test set

A test set was created by randomly extracting 20% from the dataset. The other 80% was used for training.

3 Experimental setup

To be able to predict if an apartment will be rated high/medium/low on the RentHop website 5 models were tested. For all 5 of them R was used to generate the models. The models were chosen because they are a good combination of simple linear, nonlinear and complex linear methods. Every algorithm is performed using the same data splits so the results are directly comparable.

To build the models in R the caret package[6] had to be loaded. This package streamlines model building, can be used to plot results and compare the different results. Next, all built models will be discussed shortly

3.1 Model 1: Random Forest

The random forest model is an ensemble learning method that combines multiple decision trees and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the all the individual decision trees. The model corrects for decision trees' habit of overfitting to their training set. This model is used to predict the interest level of apartments because it is very useful for predicting certain target values from earlier done observations, which is exactly what is the goal for RentHop. Using the highest accuracy and highest Kappa score R determined the ideal amount of features in the Random Forest model to be 24.

3.2 Model 2: Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is mostly used as dimensionality reduction technique in the pre-processing step before the pattern classification.The aim of the model is to project a dataset onto a lower-dimension space that has good class sperabiltity. This is to avoid overfitting and this is also called 'curse of dimensionality'. It will also reduce computational costs.[7] It is useful for categorizing apartments, since it combines features to find a pattern and looks for 2 or more classes. Therefore, Linear Discriminant Analysis is a classification model. In this case, the apartments should be ranked in interest level from low to high and therefore this model is suitable for this problem. Its purpose is to look for linear combinations of the original variables to find the best possible separation between different groups in the dataset. "LDA approaches the problem by assuming that the conditional probability density functions p(x|y=1)p(x|y=1) and p(x|y=2)p(x|y=2) are both normally distributed with mean and covariance parameters (µ1,∑1)(µ1,∑1) and (µ2,∑2)(µ2,∑2), respectively [8]". Because the standard model from the caret package was used there are no tuning parameters. A limitation of LDA is that it is sensitive to outliers [9]". Another limitation is that it cannot be used for non-linear problems and that it is likely to overfit.

3.3 Model 3: Classification & Regression Trees

Classification and regression trees are ways of predicting models from data. The models are obtained by repeated partitioning the data space and fitting a simple prediction model after each loop. Correspondingly, this can be represented as a decision tree. "Classification trees are designed for dependent variables that take a finite number of unordered values, with prediction error measured in terms of misclassification cost. Regression trees are for dependent variables that take continuous or ordered discrete values, with prediction error typically measured by the squared difference between the observed and predicted values [10] ". CART is useful for this problem, since it classifies the houses quite accurate. There have not been used specific parameters. All the standard functions of the cart() function in R are used. Also, no actions are undertaken to prune the trees. For the parameter the accuracy was used to select the optimal model using the largest value.The final value used for the model was cp = 0.005731625. There are some possible limitations of the classification and regression tree model. First, CART may have unstable decision trees. Secondly, CART only splits by one variable [11].

3.4 Model 4: K-nearest neighbors

KNN is an non-parametric lazy learning algorithm. Non-parametric means that it does not make any assumptions on the underlying data distribution. A lazy learning algorithm does not use training data or use minimal training data.Due to this fact all the data is needed for training. This can take a lot of time and memory. This model is used to predict the interest level of apartments by looking at the neighbors of an apartment and chooses the interest level that is most used by their neighbors. For the parameter accuracy was used to select the optimal model using the largest value. The final value used for the model was k = 9.The limitations of this algorithm is that it only takes the k-nearest neighbors. Only close by neighbors are taken into account, which means that maybe important apartments are not used to determine the interest level of another apartment.

3.5 Model 5: Support Vector Machines

Support Vector Machine is a supervised machine learning algorithm. It can be used for classification and regression problems. "In this algorithm, we plot each data item as a point in n-dimensional space (where n is number of features in the dataset) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyperplane that differentiate the two classes very well. Support Vectors are simply the coordinates of individual observation. Support Vector Machine is a frontier which best segregates the two classes (hyperplane/ line) [12]."For the parameter accuracy was used to select the optimal model using the largest value. The final values used for the model were sigma = 0.01495034 and Cost = 1. The limitations of this algorithm is that the best circumstances to use it is when the number of dimensions is greater than the number of samples. In the case with RentHop the number of samples is greater than the features. Also this algorithm does not perform well when having a large dataset, due to the amount of training time.

4 Results

With the models in the R environment, it was possible to test the accuracy and the Kappa score of the models running on the test set created by taking 20% of the original training set. The accuracy is the score of how much interesting levels the model predicted right. This is done by looking at the original training which provides the actual interesting level.

The other score provided by the R environment is the Kappa score. This score looks at the agreement on the actual class and the predicted class of classification The higher the score is the better the agreement is on the classification.

In the figures 3 and 4 the mean accuracy and the mean Kappa score of the five models is shown respectively. The mean is calculated by R over the 10-fold cross validations. In the figures the models represented are from left to right: *K-nearest neighbors, Linear Discriminant Analysis, Support Vector machines, Classification and Regression Trees and the Random Forest.*

When looking at the bar chart of the Kappa score and the bar chart of the accuracy one difference is notable. The Kappa score of the model *Linear Discriminant Analysis* is higher than the score of the model *Support Vector Machine*. But in the bar chart of accuracy this is not the case. This means that the accuracy and the Kappa score not always the right scores are to determine how well a model predicts a class.

One of the advantages of using R is that R can give some extra information about results. For example in the case of the model of the Random Forest, R provided the ideal amount of features to be used for the Forest. The ideal amount of features means the amount that gives the highest accuracy and highest Kappa

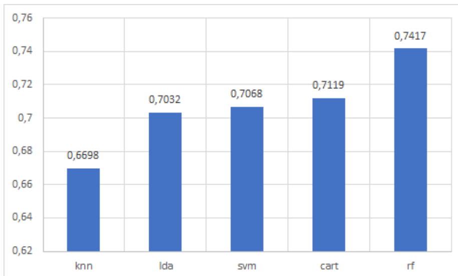

Fig. 3: Accuracy Bar chart

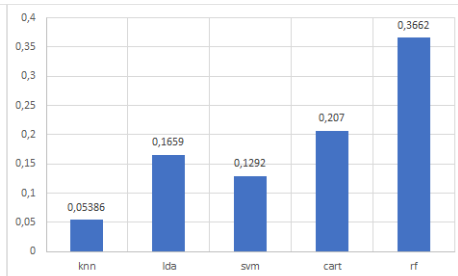

Fig. 4: Kappa Bar chart

score. For this dataset the ideal amount of features is 24. Using more features resulted in overfitting. Another feature is the ideal amount of neighbors for the K-nearest neighbors algorithm. For this dataset the ideal amount is set to 9.

When looked at both bar charts, it is clear that the model *Random Forest* is the best model to use to classify the dataset of the apartments. Although the accuracy and the Kappa score are not the best scores to have for a classification problem, it would be the best for now.

5 Discussion

The approach used for determining the best model consisted of building 5 models relatively easy using the caret package in R. Positive about this approach was that the models were built quite fast and because all algorithms used the same data splits they are easily comparable.

So far, not many limitations of this approach are detected. One small limitation of this approach is the execution time for calculating the models. Without many limitations this approach is for now the best strategy to find the best model for classifying a dataset. Something that is not considered is the classification size. In this dataset the data points were divided into 3 categories. In future this would lead to too many data points in one class. In that case it is needed to extend the categories and will the execution time expand more.

For going further with this work it would be proposed to look for a smaller execution time for the models and the exploration on more classification classes in a dataset. For the purpose of the dataset it would be useful to check which predictions are made for the provided testset on the Kaggle website with the best model of this research.

6 Conclusion

In this paper there has been an investigation for the answer to the question:

Which of the five proposed models predicts the interest level (high/medium/low) of a particular apartment best?

For this multiple algorithms are compared and analyzed. From the results you can see that by looking at both the accuracy and the Kappa score one can say that with 0,7417 and 0,3662 respectively the Random Forest model scores outstandingly better than the other 4 proposed models. This means that following this research Random Forest is the best way of predicting the interest level of an apartment on the RentHop website. The confidence interval shows for every model a value around the 0.70. The confidence intervals of all models are shown in table 2. So the conclusion can be made that every model has probably an error rate of 30%.

Table 2: Confidence Intervals

	Min Confidence	Max Confidence
knn	0,6942	0,7123
lda	0,695	0,7131
svm	0,7087	0,7265
cart	0,7032	0,7211
rf	0,7363	0,753

However as was also stated above the Random Forest scores better than the other proposed models but looking at the accuracy of Random Forest alone 0,7417 is not a very high score. It means that only 74% of the predictions made by the model are true positives and true negatives and preferably this should be much higher. For the Kappa score this is even more true because 0 would mean that all estimates done by the model are just pure luck and 1 would mean complete agreement. Therefore a score of 0,3662 is quite low.

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Table 3: Contribution table

Name	**Contribution**
Dave	Compared and created the algorithms in R
Paulien	Introduction, Abstract, Experimental setup (model 2 & 3)
Jerney	Data inspection, Experimental setup (model 4 & 5), Conclusion
Quinten	Results of the algorithms, Put the report in Latex
Pam	Data inspection, Experimental setup (model 1), Discussion

Link to the code used: https://drive.google.com/file/d/0B_-Sknri0-imYVFfRnhqVVpPWk0/view?usp=sharing